

Saarthak Saxena

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Experience

Applied AI Intern

Bruce.so Mar 2026 – May 2026

- **Pedro (Autonomous KG Agent):** Built multi-LLM autonomous agent (Gemini + Claude) maintaining **10K+** node knowledge graph with zero manual curation. Uses parallel evaluation, deferred decisions for low-confidence cases, semantic similarity for evidence ranking, and parent-child taxonomy classification.
- **Search Engine:** Rebuilt people search ranking with **LLM-generated structured constraints** (xAI/Grok), replacing fuzzy matching with deterministic exact-match gating for role, company, and location.
- **Oden (Connection Path Discovery):** Built multi-hop pathfinding across **4,739** actors and **11,609** edges. Evaluated **3.4M+** directed trials and 29.3M route combos; achieved **99.87%** success rate across 9 strategies.
- **Actor-Relationship Optimization:** Designed scoring via **60+** experiments reducing processed data to **2.93%** of baseline (72 profiles, ~90K records) while retaining **98%** high-quality edges. Cut external API calls ~60% via single-pass consolidation.
- **Knowledge Graph Visualizer:** Built full-stack interactive visualizer for exploring and debugging the live 10K+ node graph.

ML Engineer

Stealth Startup Jun 2025 – Nov 2025 Contract: Jul 2025 – Oct 2025

- **Novel GNN from Scratch:** Trained GNN from zero on novel pharmacchemistry task + dataset + model under **\$35** budget. No pretrained weights, no reference implementations, no existing benchmarks.
- **Experimentation at Scale:** Extended GraphGym for **48k** data points. Orchestrated **1,000,000+** optimization trials on distributed GPUs and beat Chemprop, the industry standard.
- **Production Delivery:** Shipped lightweight **2M**-param inference API in **8** weeks (full lifecycle: data ingestion to FastAPI + Docker + AWS deployment).
- **Business Impact:** Client used this work to secure a **INR 30 Lac (~\$35k)** research grant.

Software Engineering Intern

HeydoTech Jan 2024 – May 2024

- Developed **2** automation tools using OpenCV and PyAutoGUI for internal projects.
- Collaborated with cross-functional teams of **5+** to ship feature enhancements improving performance for **~100** users.
- Delivered UI/UX improvements with positive feedback from **80%** of test users.

Projects

Docbook Python, GraphRAG, CLI | Apr 2025 – Present

- GraphRAG-based CLI tool for intelligent documentation context extraction from single URL input.
- Converts URL to full knowledge graph using crawl4ai with automated semantic search and retrieval.

Bayesian Course Recommendation System

Python, PyTorch, Bayesian ML | 2025

- Achieved **0.915** AUC using Bayesian Personalized Ranking (BPR) for implicit feedback recommendations.
- Matrix factorization with BPR optimization and probabilistic handling for cold-start problem.

Education

BTech, Electronics and Communication Engineering Nov 2022 – Present

IIT Delhi